P1: Dynamics Derivation and Internal Verification

This P1 report establishes and verifies the reference dynamics. Whole-flight guidance and fuel optimization are outside its scope.

4 Model Force Analysis

4.1 Gravity and Standard Atmosphere

We use a local downrange–vertical plane. Position refers to the center of mass (COM), and the pitch angle points from the tail toward the nose. Gravity acts through the COM and has magnitude

$$g(h) = g_0 \left(\frac{R_E}{R_E + h} \right)^2, \qquad g_0 = 9.80665 \text{ m/s}^2, \quad R_E = 6356766 \text{ m}. \tag{1}$$

At 70 km this gives $g = 9.5942$ m/s^2, about 2.17% below g_0. This is a spherical, nonrotating reference approximation, not a latitude-resolved gravity model.

Atmospheric properties follow the lower layers of the 1976 standard atmosphere [3]. Geometric altitude h is converted to geopotential altitude $H = R_E h/(R_E + h)$. In a layer with base (H_b, T_b, p_b) and temperature gradient λ,

$$T_a = T_b + \lambda(H - H_b), \tag{2}$$

$$p_a = \begin{cases} p_b (T_a/T_b)^{-g_0/(R_a \lambda)}, & \lambda \neq 0, \\ p_b \exp[-g_0(H - H_b)/(R_a T_b)], & \lambda = 0, \end{cases} \tag{3}$$

$$\rho = \frac{p_a}{R_a T_a}, \qquad R_a = 287.05287 \text{ J/(kg K)}. \tag{4}$$

These expressions follow from hydrostatic balance $dp_a/dH = -\rho g_0$ and the ideal-gas law. The implemented range ends at $H = 84.852$ km; trajectories outside that range are rejected rather than assigned extrapolated density.

4.2 Axial and Crossflow Aerodynamics

The cylindrical approximation gives $A_{end} = \pi D^2/4 = 38.4845$ m^2 and side area $DL = 455$ m^2. Define body directions

$$\mathbf{e} = (\cos\theta, \sin\theta), \quad \mathbf{n} = (-\sin\theta, \cos\theta), \quad \mathbf{u} = \mathbf{v} - \mathbf{w}, \quad u_a = \mathbf{u} \cdot \mathbf{e}, \quad u_n = \mathbf{u} \cdot \mathbf{n}. \tag{5}$$

Instead of applying one coefficient to a changing projected area, we separate axial and crossflow contributions. Cylinder research motivates treating crossflow as a distinct aerodynamic regime [2]; our constant coefficients are low-order engineering assumptions, not a fit to that report.

The axial force is

$$F_a = -\tfrac{1}{2} \rho C_a A_{end} u_a |u_a|. \tag{6}$$

For an equivalent crossflow strip at signed axial arm s, rotation adds the local normal velocity ωs. Thus

$$dF_n = -\tfrac{1}{2} \rho C_n D (u_n + \omega s) |u_n + \omega s| \, ds. \tag{7}$$

With N midpoint strips of length $\Delta s = L/N$ and $s_i = -L/2 + (i - 1/2)\Delta s + \Delta l$,

$$\mathbf{F}_D = F_a \mathbf{e} + \sum_i F_{n,i} \mathbf{n}, \qquad M_a = \sum_i s_i F_{n,i}. \tag{8}$$

The shifted equivalent distribution represents an uncertain pressure-center offset Δl; it is not a physical surface mesh. We evaluate density at the COM and neglect its variation across the 65 m body. Axial force acts along the centerline. Figure 1 defines the forces and signs.

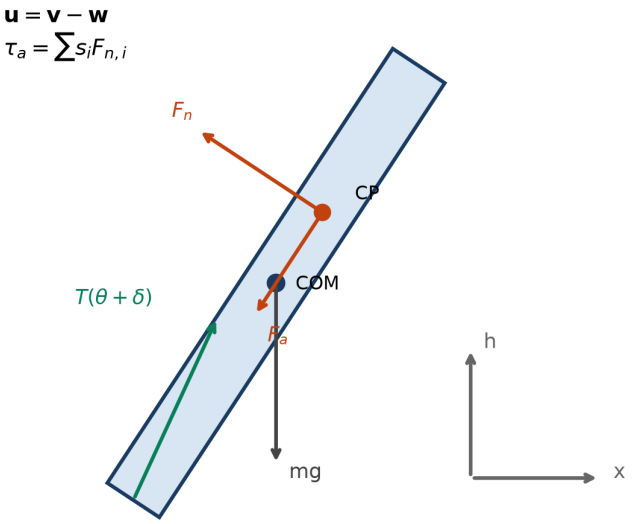
For zero rotation, $M_a = \Delta l F_n$. Consequently, even spatially uniform wind can generate a moment when $\Delta l \neq 0$. For a centered distribution and pure rotation, direct integration gives

$$M_a = -\frac{\rho C_n D L^4}{64} \omega |\omega|. \tag{9}$$

This opposes rotation. More generally, the mechanical power transferred by the modeled air-relative forces satisfies

$$\mathbf{F}_D \cdot \mathbf{u} + M_a \omega = -\tfrac{1}{2} \rho C_a A_{end} |u_a|^3 - \tfrac{1}{2} \rho C_n D \sum_i \Delta s |u_n + \omega s_i|^3 \leq 0. \tag{10}$$

This identity provides a useful sign and implementation check. It concerns motion relative to air, not ground-frame energy in moving wind.



Rigid cylinder; signed pitch measured from +x.
CP offset and gimbal angle exaggerated.

Figure 1: Planar reference geometry and force conventions. The pressure-center displacement is exaggerated. Positive gimbal angle at the tail produces a negative pitch moment.

4.3 Reference Propulsion Parameters

The maximum landing thrust is set to 2.846 MN by analogy with the published BE-4 thrust scale [1]. We select effective $I_{sp} = 330$ s, oxygen-to-methane mass ratio 3.5, minimum throttle 0.4, and gimbal limit 10° as reference-design assumptions. The mixture-ratio choice is informed by a methane/oxygen mission study [4]; that study's vacuum

specific impulse is not assigned to this landing engine. We set $C_a = 0.8$, $C_n = 1.2$, and $\Delta l = 0.05L$. Parameter ranges are specified in the P0 reference contract and must be explored before a final fuel recommendation.

The standard effective-exhaust-velocity definition gives $c = I_{sp} g_0$ and

$$\dot{m} = -T/c, \qquad T = 0 \quad \text{or} \quad 0.4T_{max} \le T \le T_{max}. \tag{11}$$

This consumes both oxygen and methane. Pressure dependence and actuator transients are represented only through the effective parameter assumptions at this stage.

5 Model Motion Analysis

5.1 Momentum Balance and Pitch Dynamics

The state is $\mathbf{y} = (x, h, v_x, v_h, m, \theta, \omega)$. Accounting for exhaust momentum through thrust, the open-system translational balance is $m\dot{\mathbf{v}} = m\mathbf{g} + \mathbf{F}_D + \mathbf{F}_T$. One must not add a second $\dot{m}\mathbf{v}$ term to this thrust-based equation. The coupled model is

$$\dot{x} = v_x, \qquad \dot{h} = v_h, \tag{12}$$

$$\dot{v}_x = \frac{F_{D,x} + T\cos(\theta + \delta)}{m}, \qquad \dot{v}_h = -g(h) + \frac{F_{D,h} + T\sin(\theta + \delta)}{m}, \tag{13}$$

$$\dot{m} = -\frac{T}{g_0 I_{sp}}, \qquad \dot{\theta} = \omega, \tag{14}$$

$$I_d\dot{\omega} = M_a - \frac{L}{2}T\sin\delta, \qquad I_d = \frac{m_d(3R_b^2 + L^2)}{12}. \tag{15}$$

The thrust application point is $-(L/2)\mathbf{e}$, so its cross product with thrust gives the negative sign above. The dry body is modeled as a uniform cylinder. Remaining propellant is idealized as concentrated at the COM with negligible pitch inertia and no additional rotational angular-momentum flux. Hence I_d is constant. A distributed-tank model would require consistent treatment of changing inertia and exhaust angular momentum; the present equation does not claim to describe those effects.

5.2 Initial Conditions and Contact Geometry

For each selected initial flight-path angle,

$$x_0 = 0, \quad h_0 = 70000 \text{ m}, \quad (v_{x,0}, v_{h,0}) = \frac{6000}{3.6}(\cos\gamma_0, \sin\gamma_0), \quad m_0 = m_d + m_{p,0}. \tag{16}$$

The reference uses $\gamma_0 = \theta_0 = 15^\circ$ and $\omega_0 = 0$. A trial $m_{p,0} = 80000$ kg initializes calculations; it is not a computed minimum.

For a cylindrical body with no landing gear, the vertical support extent below the COM is $(L/2)|\sin\theta| + R_b|\cos\theta|$. First deck contact therefore occurs when

$$f(\mathbf{y}) = h - h_{deck} - \frac{L}{2}|\sin\theta| - R_b|\cos\theta| = 0. \tag{17}$$

Only an upright body contacts at $h - h_{deck} = L/2$. Speed, attitude, pitch-rate, propellant reserve, and position relative to the chosen ship must be assessed separately. The local diagnostic below checks the former quantities, not fixed-ship position capture.

5.3 Analytic Baselines and Scale Estimates

For constant gravity, zero thrust, and zero drag,

$$h(t) = h_0 + v_{h,0}t - \tfrac{1}{2}g_0t^2, \qquad x(t) = v_{x,0}t. \tag{18}$$

At 15°, $(v_{x,0}, v_{h,0}) = (1609.88, 431.37)$ m/s. The constant-g_0 apex estimate is 79.49 km after 43.99 s. This is an analytic scale estimate, not the integrated reference trajectory. In the variable-gravity vertical vacuum limit, the conserved specific energy is

$$E = \tfrac{1}{2}v_h^2 - \frac{g_0R_E^2}{R_E + h}. \tag{19}$$

Without opposing external acceleration in the thrust direction, integrating $dv = -c\,dm/m$ yields

$$\Delta v = c\ln(m_0/m_f). \tag{20}$$

For 210 t initially and 135 t finally, this gives 1429.86 m/s. It is ideal propulsive capability, not a required landing velocity increment or proof of feasibility. Atmospheric drag may remove speed while gravity and steering alter the propulsion requirement.

With mass frozen, no drag, and vertical maximum thrust, a useful stopping-distance estimate is

$$d_{stop} \simeq \frac{v_h^2}{2(T_{max}/m - g_0)}, \qquad T_{max} > mg_0. \tag{21}$$

For 180 t and a 90 m/s descent, it is 674.5 m. Mass loss and drag are omitted from this estimate. The maximum-thrust weight ratio is 1.382 at 210 t and 2.232 at dry mass, showing why instantaneous mass must enter the dynamics.

5.4 Numerical Integration and Local Powered-Braking Check

A fixed-step fourth-order Runge–Kutta integrator advances the coupled equations. Step size is reduced near the deck, and contact is located by linear interpolation across a sign change of Eq. (17). This does not imply fourth-order event accuracy. A conservative propellant guard stops before a step could cross dry mass; it prevents negative inventory but does not return the exact depletion time.

To test the powered and rotational implementation independently of whole-flight optimization, we start a local case at $h = 1200$ m, $v_h = -90$ m/s, $v_x = 0$, and $m = 180000$ kg. Define height above upright contact $z = h - L/2$ and a descent-speed reference

$$v_d = -\min\{90, \sqrt{2k_z\max(z, 0.01) + v_c^2}\}, \quad k_z = 1.5\ \text{m/s}^2, \quad v_c = 1\ \text{m/s}. \tag{22}$$

A vertical acceleration command $a_c = k_v(v_d - v_h)$, $k_v = 4\ \text{s}^{-1}$, supplies the throttle request $[m(g_0 + a_c) - F_{D,h}]/T_{max}$, clipped to the permitted operating range. This vertical-axis test controller uses g_0 as an approximation and does not provide general two-dimensional guidance. Pitch control requests

$$M_{req} = I_d[k_p(\pi/2 - \theta) - k_d\omega] - M_a, \quad \delta = \arcsin\left[\text{clip}\left(-\frac{M_{req}}{(L/2)T}, -1, 1\right)\right], \tag{23}$$

where $k_p = 0.5\ \text{s}^{-2}$, $k_d = 1.2\ \text{s}^{-1}$, followed by gimbal clipping. With no saturation, this gives the stable error equation $\ddot{e} + k_d\dot{e} + k_pe = 0$. Saturation and engine-off motion invalidate that linear closed-loop conclusion and must be checked in the full trajectory.

At a 0.005 s step the nominal case contacts after 35.97505 s with $v_h = -1.50291$ m/s and consumes 22679.12 kg. A separate test with initial pitch 89° and initial rate 0.005 rad/s ends with approximately zero pitch error and $v_x = 0.4671$ m/s. Its horizontal drift is 16.56 m, exceeding the 5 m fixed-ship position tolerance if the ship remains at $x = 0$. Thus attitude recovery is demonstrated locally, but fixed-ship capture is not. Figure 2 shows these implementation checks; they do not estimate separation fuel demand.

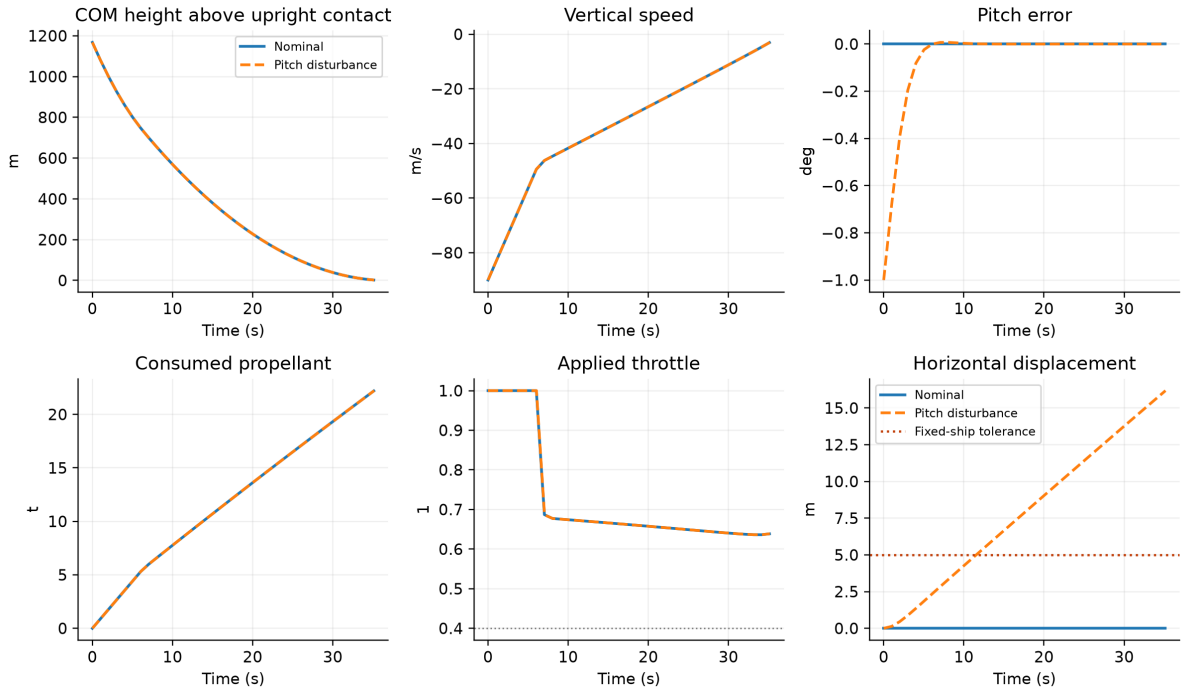


Figure 2: Local powered-braking verification, not a separation-to-landing solution. The disturbed case starts at 89° pitch with 0.005 rad/s pitch rate. The 5 m line highlights that position capture is not guaranteed by attitude recovery.

6 Internal Verification

6.1 P1 Internal Verification and Numerical Convergence

The verification suite tests implementation consistency and known limiting solutions; no flight-data validation is claimed. Standard-atmosphere base pressures at 0, 11, 20, 32, and 47 km geopotential altitude agree with rounded reference values within 4.8×10^{-6} relative error. Out-of-range atmospheric queries are rejected. A vertical vacuum coast conserves the modeled specific mechanical energy to 7.5×10^{-9} J/kg over 20 s. Constant-thrust mass history and horizontal ideal-rocket velocity agree with their analytic expressions to 7.6×10^{-10} kg and 2.9×10^{-14} m/s, respectively, in the chosen tests. These small residuals apply to those tests and do not measure real-world accuracy.

Pure axial and pure crossflow force limits, static pressure-center torque, torque-free rotation, damping sign, gimbal sign, and air-relative dissipativity are checked. For pure centered rotation, Eq. (9) provides an independent integral. Midpoint-strip relative errors are 2%, 0.5%, 0.125%, 0.03125%, and 0.0078125% for 10, 20, 40, 80, and 160 strips. Thus the default 20-strip rotational torque has a 0.5% quadrature error in this limiting case; this is distinct from uncertainty in the aerodynamic coefficients.

A four-second smooth unpowered trajectory is compared with a 0.00125 s reference using the same equations. Position differences decrease from 9.75×10^{-7} m at 0.08 s to 2.76×10^{-10} m at 0.01 s. Since force laws contain absolute values and feedback contains clipping, fourth-order behavior is not assumed across every nonsmooth transition. Figure 3 separates smooth integration, strip quadrature, and touchdown-event checks. Between the final two steps, consumption changes by 0.00384 kg and contact

Table 1: Local terminal-event convergence. All rows satisfy the selected speed, attitude, and reserve criteria; this table does not test fixed-ship position capture.

Step (s)	Contact time (s)	v_h (m/s)	Consumed (kg)
0.080	35.975945	-1.502940	22679.544
0.040	35.975286	-1.503756	22679.191
0.020	35.975097	-1.502978	22679.139
0.010	35.975063	-1.502986	22679.123
0.005	35.975047	-1.502909	22679.119

time by 1.60×10^{-5} s. Event speed differences are not monotonic, so no fourth-order contact-convergence claim is made. The fuel-depletion guard is verified to avoid dry-mass overshoot, while exact depletion-time localization remains outside this conservative guard's contract. The reference implementation and P1 checks are reproducible with `node src/verify-p1.mjs`. The remaining limitations are model-form uncertainty, actuator idealization, atmosphere range, local-plane/rotation approximations, and absent structural/thermal allowables. Whole-flight feasibility and fuel optimization are delegated to the next stage.

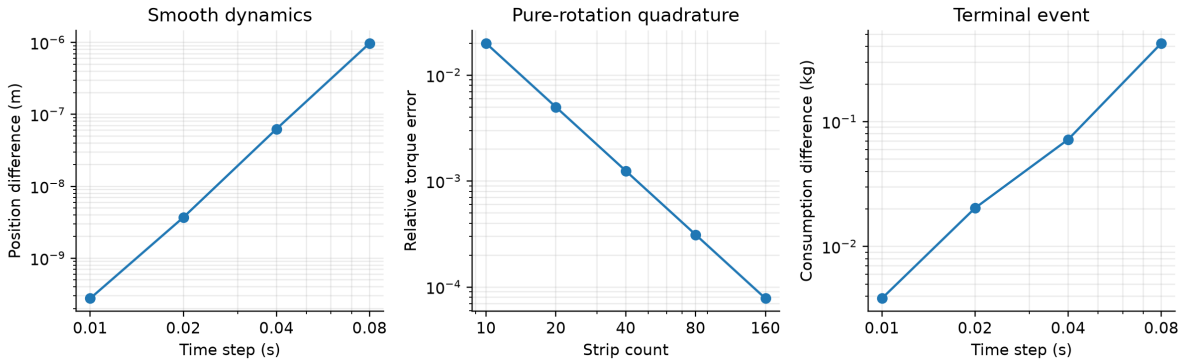


Figure 3: Independent numerical error sources. Left: smooth-trajectory position difference relative to a finer same-model integration. Middle: strip torque error against the analytic integral. Right: contact-consumption difference relative to the 0.005 s run.

References

[1] Blue Origin. BE-4. Manufacturer webpage, accessed November 7, 2026. https://www.blueorigin.com/es-MX/engines/be-4.

[2] F. E. Gowen and E. W. Perkins. Drag of Circular Cylinders for a Wide Range of Reynolds Numbers and Mach Numbers. NACA-TN-2960, 1953.

[3] NOAA, NASA, and USAF. U.S. Standard Atmosphere, 1976. NASA-TM-X-74335, 1976.

[4] Mars Sample Return Mission Utilizing In-Situ Propellant Production. NASA NTRS record 19950021840. Methane/oxygen performance discussion, Fig. 2.

[5] UPC 2026, Problem A: Landing a First-Stage Rocket Booster. User-provided problem image.